

Madachy Recent 2026 Publications

Book Chapters

R. Madachy, R. Bell, and R. Longshore, "A Generative AI-driven Systems Engineering Maturity and Cost Modeling Framework," in Proceedings of the 22nd Annual Conference on Systems Engineering Research (CSER), A. M. Madni, D. Verma, M. Sievers, M. Wheaton, E. Ordoukhanian, and S. Purohit, Eds. Cham, Switzerland: Springer, 2026

Refereed Journal Articles

R. Bell, R. Madachy, and R. Longshore, " *Balancing Accuracy and Cost: Trade-Offs in Large Language Model Quantization for the Systems Engineering Domain.*" *Systems Engineering* (2026): e70061. <https://doi.org/10.1002/sys.70061>

R. Bell, R. Longshore, R. Madachy, and A. Hanrahan, " *Baselining Large Language Model Performance in Systems Engineering Using SysEngBench.*" *Systems Engineering* (2026): e70067. <https://doi.org/10.1002/sys.70067>

Refereed Magazine and Practitioner Articles

R. Madachy. Trustworthy AI for quantitative systems engineering via deterministic delegation to verified tools. *INCOSE INSIGHT*, 29(3):58-66, June 2026.

R. Madachy, D. O'Leary, T. McDermott, and S. Bank. Twenty tips for using generative AI in systems engineering. *INCOSE INSIGHT*, 29(3):29-31, June 2026.

Conference Papers

R. Madachy, A Personal Generation Process Model for Human – AI Collaboration in Systems Engineering, in Proceedings of the 23rd Annual Conference on Systems Engineering Research (CSER), 2026

Conference Presentations and Tutorials

R. Madachy, and R. Bell, Tutorial - Developing AI Agents and Custom LLMs for Systems Engineering, in Proceedings of the 23rd Annual Conference on Systems Engineering Research (CSER), 2026

R. Madachy, O. Yakimenko, J. Sweeney, R. Giachetti, and M. Rhoades, Enhancing Systems Engineering Academic Processes with AI at NPS, in Proceedings of the 23rd Annual Conference on Systems Engineering Research (CSER), 2026

R. Madachy, Architecting Trustworthy AI for Systems Engineering Cost Estimation, In Proceedings of the 36th International Symposium of INCOSE, 2026

R. Madachy, Understanding and Improving Human–AI Collaboration in Systems Engineering through Simulation, In Proceedings of the 36th International Symposium of INCOSE, 2026

R. Bell, and R. Madachy, An Empirical Meta Evaluation of Large Language Model Evaluation Methods for Systems Engineering, In Proceedings of the 36th International Symposium of INCOSE, 2026